C++ Interview Preparation Guide

1. Object-Oriented Programming (OOP)

- Classes and Objects
  - Constructors and Destructors (Default, Parameterized, Copy)
  - Friend Functions and Classes
- Four Pillars of OOP: Encapsulation, Abstraction, Inheritance, Polymorphism
- Virtual Polymorphism: vtable, vpointer
- Function Overloading and Operator Overloading
- Function Overriding
- Trick Question: Create one class and one object such that code compiles but object is not created (Pure Virtual Function)
- Abstract Class and Interface
  - Pure Virtual Functions vs. Virtual Functions
- RTTI (Run-Time Type Information)
  - typeid and dynamic_cast
- Upcasting vs Downcasting (Which is safe? → Dynamic Downcasting)
- Object Slicing
- Lambda Expressions
  - Captures (by value, by reference)
  - Mutable Lambdas

2. Pointers

- Smart Pointers: unique_ptr, shared_ptr, weak_ptr
  - make_unique and make_shared
  - Cyclic References with shared_ptr
- RAII (Resource Acquisition Is Initialization)
- Rule of 0/3/5
- Function Pointers
  - Member Function Pointers
- Bitwise Operations:
  - Set a bit, Unset a bit, Toggle a bit
  - Check if a number is a power of two
  - Count number of set bits (Brian Kernighan's Algorithm)
  - Reverse bits
  - Convert int to binary
  - Find the lowest/highest set bit
  - Bit manipulation for optimization (e.g., masks)
- Structure Padding and Packing
  - #pragma pack and alignment

3. Storage Classes

- auto, register, static, extern
  - Thread-local storage (thread_local)
  - Volatile keyword

4. Data Structures

a. Vectors

- Remove duplicates (sorted and unsorted)
- Print non-repeating elements
- Merge two sorted arrays
- Two-sum approach (sum = target)
- Rotate image or 2D arrays
  - Transpose matrix
  - Spiral traversal

b. Arrays

- Basic operations and algorithms
  - Kadane's Algorithm for maximum subarray sum
  - Dutch National Flag (sort 0s, 1s, 2s)
  - Majority element (Moore's Voting Algorithm)

c. Strings

- Find non-repeating character
- Reverse and check palindrome (string or sentence)
- Check if string is an anagram
  - Longest common substring/subsequence
  - String matching algorithms (KMP, Rabin-Karp)
  - Substring search

d. Linked List

- Insert/Delete a node
- Insert/Delete nth node
- Reverse linked list (iterative and recursive)
- Check if linked list is palindrome
- Merge two sorted lists
- Detect cycle in linked list (Floyd's Cycle Detection)
  - Find middle node
  - Remove duplicates from sorted/unsorted list
  - Intersection of two linked lists
  - Clone linked list with random pointers

e. Dynamic Programming

- House Robber problem
- Find missing number in an array
- Recursion algorithms
- Fibonacci and Factorial (memoization, tabulation)
  - 0/1 Knapsack
  - Longest Increasing Subsequence (LIS)
  - Coin Change problem
  - Matrix Chain Multiplication
  - Edit Distance

Operating Systems (OS)

1. Multithreading

- Mutex vs Semaphores
  - Binary vs Counting Semaphores
- Deadlocks and Prevention (Banker's Algorithm)
- Producer-Consumer Problem
- Create threads and print:
  - Even and odd numbers alternately
  - Sequence: 1,2,3 repeated (e.g., ABC, ABC, ABC…)
- Binary Semaphores (Acquire and Release Resource)
- Leveraging Lambda Expressions in Multithreading
  - Thread pools
  - Condition Variables
  - Race Conditions and Critical Sections
  - std::future and std::promise

2. QNX (Adaptive AUTOSAR)

- Hypervisor Concepts and How Hypervisor Works
  - Type 1 vs Type 2 Hypervisors
  - Virtualization in Automotive (e.g., AUTOSAR Adaptive Platform)
  - Partitioning and Isolation

FAQ

- Difference between Macro and Inline Functions
- Difference between const and constexpr
- Compilation Process (Preprocessing, Compilation, Assembly, Linking)
- Memory Layout: Stack, Heap, Data Segment, Code Segment (with storage class info)
- Difference between static and const
    - constexpr vs consteval (C++20)
    - Linkage (internal vs external)

Communication Protocols

- CAN, CAN FD, Extended CAN
    - Arbitration and Error Handling in CAN
- Ethernet
    - VLANs and Switching
- I2C
    - Master-Slave Communication, Clock Stretching
- UART
    - Baud Rate, Parity, Flow Control
- CAN DBC
    - Signal Encoding/Decoding
- DEM (Diagnostic Event Manager)
- DIAG (Diagnostics)
    - UDS (Unified Diagnostic Services)
- LIN Protocol (Automotive)
    - Master-Slave Architecture
- Basics of TCP/IP vs UDP
    - Three-Way Handshake, Congestion Control

Extra Topic Suggestions

C++

- Move Semantics and std::move
    - Rvalue References
- Copy Constructor vs Move Constructor
    - Assignment Operator (Copy vs Move)
- Exception Handling (try, catch, throw)
    - noexcept specifier
    - Stack Unwinding
- Templates (Function and Class Templates)
    - Template Specialization
    - Variadic Templates
- STL (Standard Template Library): map, set, unordered_map
    - Iterators and Algorithms (std::sort, std::find)
    - Priority Queue (heap)
    - Deque and List
- Memory Management: new, delete, malloc, free
    - Placement new
    - Memory Leaks and Tools (Valgrind)
- Thread Safety and Atomic Operations
    - std::atomic and Memory Models

OS

- Scheduling Algorithms (Round Robin, Priority Scheduling)
    - FCFS, SJF, Multilevel Queue
- Inter-Process Communication (IPC)
    - Pipes, Message Queues, Shared Memory, Signals
- Virtual Memory and Paging
    - Page Replacement Algorithms (LRU, FIFO)
    - Thrashing and Demand Paging

Communication

- Basics of TCP/IP vs UDP
- LIN Protocol (Automotive)